

Thinking Ahead to a Humid Summer



ALAN MEIER

John Proctor is the president of Proctor Engineering Group in San Rafael, California.

Q. I recently read that a variable speed, two stage air conditioner can maintain 50% relative humidity (RH) even when the latent load is 60% of the total load. Would this really work?

—Skeptic from the Rot Belt

A. What you heard exaggerates the true capability of a two-speed compressor, variable-speed fan machine. I know of no residential air conditioner made that can accomplish 60% latent cooling (moisture removal) and 40% sensible cooling with a 50% indoor RH.

Using a Carrier 38TDB060-31 with an FV4BNB006 indoor section for example, the absolute best you will get at low speed (300 CFM per ton) is 23% latent cooling (77% sensible heat ratio). That occurs only in the following unlikely scenario: 75°F and 50% RH inside, 75°F outside, running continuously, with no heat gains in the return duct system.

The way latent cooling increases with conventional A/C (including a two-speed machine) is that the indoor RH rises above 50%. The indoor RH rises until there is a balance between the latent capacity and the latent load. For example, at the same conditions as above except that the indoor RH has risen to 66%, the unit produces a 61% sensible heat ratio (39% latent capacity).

Simply speaking, a conventional air conditioner is not a great dehumidifier. This is because no condensation is formed and no moisture removed until the return air reaches its dew point. All of the cooling to the dew point is sen-



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sible cooling and uses a good deal of the total capacity available from the machine. The amount of sensible cooling to the dew point increases as the return plenum RH drops. So if we are going to maintain a 50% indoor RH, there is lot of sensible cooling that will take place.

One can reduce the amount of sensible cooling by reducing the air flow. This is partially explained by the relationship among sensible cooling (Btu•h), temperature drop (°F), and air flow (CFM). Cooling Btu•h = 1.08 x change in °F x CFM. The dew point for 75°F / 50% RH air is 55°F. At 55°F, moisture will begin to condense out of the air. The °F is 75 - 55 = 20°F. If we are moving 1,000 CFM through the unit, it will take

21,600 Btu•h to bring the air to the dew point (1.08 x 20 x 1,000). If we move 700 CFM through the unit, it will take 15,120 Btu•h to bring the air to the dew point. The extra 6,480 Btu•h will be used to remove moisture and further reduce the supply temperature.

A variable-speed fan on a two-speed air conditioner slows down to remove more moisture. There is a practical limit to reduced air flow—too low and you get an iced coil.

Incidentally, a conventional dehumidifier is not really much better. It too has to bring the return air down to the dew point (using its capacity). What a dehumidifier does is dump the heat it removes (and then some) back

into the indoor air. This negates the sensible cooling.

The best way to maintain comfort in a climate that has high summer humidity is to reduce the infiltration of moist outdoor air into the house with a well-sealed building shell and duct system. This means you will be able to control (and potentially dehumidify) the air ventilating the home.

The second thing to do is to be sure that the return ducts are not picking up heat between the return grilles and the return plenum.

In these humid climates you will probably have to run your A/C fan at a lower speed than what is used in Arizona, and if that is not enough, add a dehumidifier.

